SYSDESIGN REFERENCE

SQL vs NoSQL Interview Drill

A timed prompt, probing questions, rubric, and answer outline.

Select a data model and consistency model from access patterns, not product labels.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / SQL-VS-NOSQL

SYSDESIGN

SQL vs NoSQL Interview Drill

CORE IDEA

Select a data model and consistency model from access patterns, not product labels.

Reference architecture decision chain

01 Transactions	02 Joins	03 Schema Evolution	04 Scale Shape

Decisions to defend

+ Transactions
+ joins
+ schema evolution
+ scale shape
+ indexing
+ and operational maturity.

Failure modes to test

+ Polyglot sprawl
+ denormalization bugs
+ and choosing for hypothetical scale.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

SQL vs NoSQL Interview Drill

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS

REVIEW GATE: TECHNICAL REVIEW. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.